

AMP Phase 1 scRNA-seq Analysis — Methods for Statistician Review

Project: Transcriptional Dysfunction of FOXP3+ Tregs in RA Synovium
Date: 2026-06-08

1. Dataset

Property	Value
Study	Zhang et al. 2019, <i>Nat Immunol</i> 20:928–942
Repository	Broad Single Cell Portal, accession SCP279
Also deposited at	ImmPort SDY998
Technology	CEL-Seq2, 384-well Smart-seq plates
Total cells	5,265 synovial cells
Donors	25 (4 OA, 21 RA) — disease labels not mappable to individual cells without ImmPort metadata
Expression units	Log ₂ -normalised CPM, pre-normalised by data depositors
Data access	SCP public REST API: GET https://singlecell.broadinstitute.org/single_cell/api/v1/studies/SCP279/clusters/t-SNE%20coordinates%20RA?gene={GENE} — returns per-cell expression for one gene at a time
Genes retrieved	113 total (97 original signature genes + 16 HIF genes), fetched individually

2. T Cell Clusters

Cluster	N cells	Identity
SC-T1	635	CCR7+ CD4+ T cells (naive/central memory)
SC-T2	48	FOXP3+ Tregs
SC-T3	70	PD-1+ Tph/Tfh cells

Cluster	N cells	Identity
SC-T4	603	GZMK+ CD8+ T cells
SC-T5	63	GNLY+ GZMB+ CTLs
SC-T6	110	GZMK+/GZMB+ mixed CD8+ T cells

Cluster labels retrieved from the same SCP API endpoint (`data.annotations` field).

3. Gene Signatures

Treg Identity (11 genes)

FOXP3, IL2RA, CTLA4, TIGIT, TNFRSF9, ENTPD1, PRDM1, TNFRSF18, DUSP4, NR4A2, IKZF2

STING / IFN (27 genes)

CGAS, TBK1, IRF3, IRF7, IRF9, STAT1, STAT2, MX1, MX2, IFIT1, IFIT2, IFIT3, ISG15, ISG20, OAS1, OAS2, OAS3, IFI44, IFI44L, IFITM1, IFITM3, DDX58, IFIH1, RSAD2, CXCL9, CXCL10, TMEM173

ROS / Oxidative Stress (24 genes)

SOD1, SOD2, CAT, GPX1, GPX4, PRDX1, PRDX2, PRDX3, PRDX5, TXNRD1, TXNRD2, NQO1, HMOX1, GCLC, GCLM, SLC7A11, NFE2L2, KEAP1, SRXN1, SESN2, SQSTM1, NCF1, NCF2, CYBB

Heat Stress / UPR (22 genes)

HSPA1A, HSPA1B, HSPA5, HSPA8, HSPB1, HSPD1, HSPE1, HSPH1, HSP90AA1, HSP90AB1, DNAJB9, DNAJC3, CALR, PDIA4, HYOU1, XBP1, ATF4, ATF6, EIF2AK3, ERN1, DDIT3, HERPUD1

HIF / Hypoxia (16 genes)

HIF1A, EGLN3, LDHA, SLC2A1, PKM, PGK1, ENO1, BNIP3, BNIP3L, DDIT4, VEGFA, PDK1, PFKFB3, SLC16A3, CD274, CXCR4

Source: Lombardi et al. 2022 Cell Reports (consensus HIF target set); Dang et al. 2011 Cell; Hsu et al. 2020 Nat Commun. Added later in the session — no formal statistical test was run for this pathway.

4. Pathway Scoring

Method: Mean log₂-normalised expression of signature genes per cell.

```
score(cell, pathway) = mean( expression[gene_i] for gene_i in signature )
```

- Expression values are log₂-normalised CPM as deposited; no additional transformation applied
- All 113 genes were present in the dataset (VSIR and CGAS retrieved via aliases C10orf54 and MB21D1 respectively)

5. Statistical Testing

- **Comparison:** SC-T2 Tregs (n=48) vs all other T cell clusters combined (n=1,481)
- **Test:** Two-sided Wilcoxon rank-sum test (`scipy.stats.mannwhitneyu, alternative='two-sided'`)
- **Multiple testing correction:** Benjamini-Hochberg FDR across 4 pathway comparisons (Treg Identity excluded as confirmatory; HIF not tested)
- **Effect size:** Rank-biserial correlation $r = 1 - (2U / (n_1 \times n_2))$

Results:

Pathway	SC-T2 mean	SC-T2 median	Other T mean	Other T median	p-value	FDR adj-p	Effect size r
Treg Identity	5.285	5.033	1.602	1.313	1.9×10^{-27}	(confirmatory)	0.919
STING / IFN	1.890	1.750	1.282	1.187	9.6×10^{-6}	3.8×10^{-5}	0.375
ROS / Oxidative	1.745	1.661	1.494	1.428	0.026	0.053	0.188
Heat Stress / UPR	4.213	4.255	3.923	3.881	0.119	0.159	0.132
Exhaustion / Checkpoint	2.268	2.001	2.397	2.156	0.299	0.299	0.088

Full statistics (including exact U values) are in `pathway_score_statistics.csv`.

6. Heatmap Methods (fig3, fig3b)

- **Input:** Per-cell log₂-CPM expression from SCP279 API
- **Aggregation:** Mean expression per cluster across SC-T1, SC-T2, SC-T3
- **Z-scoring:** Per gene across the 3 cluster means:

$$z = (\text{mean_cluster} - \text{mean_across_3_clusters}) / \text{std_across_3_clusters}$$

- **Clipping:** ± 1.15 (this is the mathematical maximum with 3 groups, not an arbitrary threshold)
- **Color scale:** RdBu_r diverging colormap

- **fig3:** 5 pathways, 100 genes (includes Heat Stress/UPR, excludes Exhaustion/Checkpoint)
 - **fig3b:** 4 pathways, 78 genes (excludes Heat Stress/UPR and Exhaustion/Checkpoint)
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7. Key Limitations

1. **Small Treg N (n=48).** All findings are exploratory. The STING/IFN result ($\text{FDR}=3.8\times 10^{-5}$, $r=0.375$) is robust; ROS ($\text{FDR}=0.053$) is a trend only.
2. **No disease-level comparison.** Plate-to-patient disease mapping requires ImmPort SDY998 metadata (login required); not available for this analysis.
3. **Pathway scoring = simple mean.** Does not account for gene-gene correlations or background expression. Scanpy `score_genes()` or GSVA would be more rigorous.
4. **HIF pathway not formally tested.** Added descriptively; Wilcoxon test not run.
5. **No batch correction.** Data used as deposited from SCP279.
6. **CEL-Seq2 dropout.** Low-expressed genes may be underestimated relative to 10x Chromium data.